

Math refresher

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \langle u, v \rangle = u_1 v_1 + u_2 v_2 = (u, u_2) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = u^T v$$

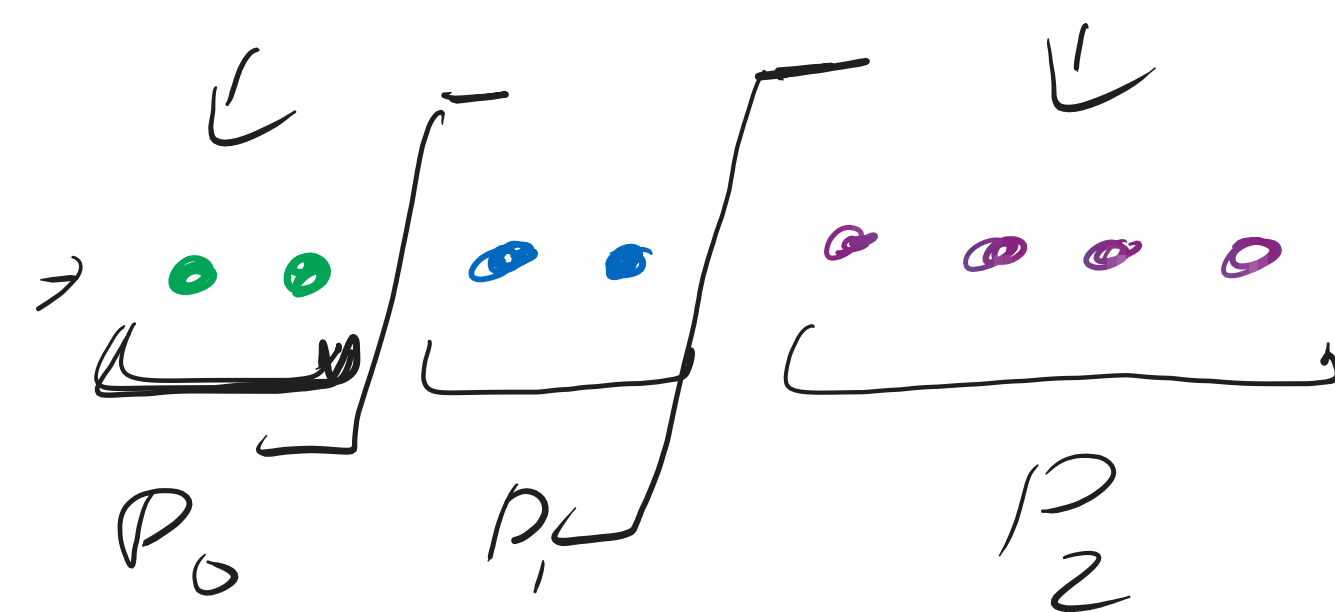
$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -2_1 & - \\ -2_2 & - \end{bmatrix}; \quad Au = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} a \cdot u_1 + b \cdot u_2 \\ c \cdot u_1 + d \cdot u_2 \end{bmatrix} = \begin{bmatrix} -2_1 & - \\ -2_2 & - \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} \langle 2_1, u \rangle \\ \langle 2_2, u \rangle \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ c_1 & c_2 \end{bmatrix}; \quad AB = \begin{bmatrix} 1 & 1 \\ c_1 & c_2 \end{bmatrix} = \begin{bmatrix} \langle 2_1, c_1 \rangle & \langle 2_1, c_2 \rangle \\ \langle 2_2, c_1 \rangle & \langle 2_2, c_2 \rangle \end{bmatrix}$$

$$\begin{aligned} (Au)^T &= \left(\begin{bmatrix} -2_1 & - \\ -2_2 & - \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right)^T = \left(\begin{bmatrix} \langle 2_1, u \rangle \\ \langle 2_2, u \rangle \end{bmatrix} \right)^T = \\ &= u^T A^T = \begin{pmatrix} -2_1 & -2_2 \end{pmatrix} \begin{bmatrix} 1 & 1 \\ c_1 & c_2 \end{bmatrix} \end{aligned}$$

Cross-Entropy

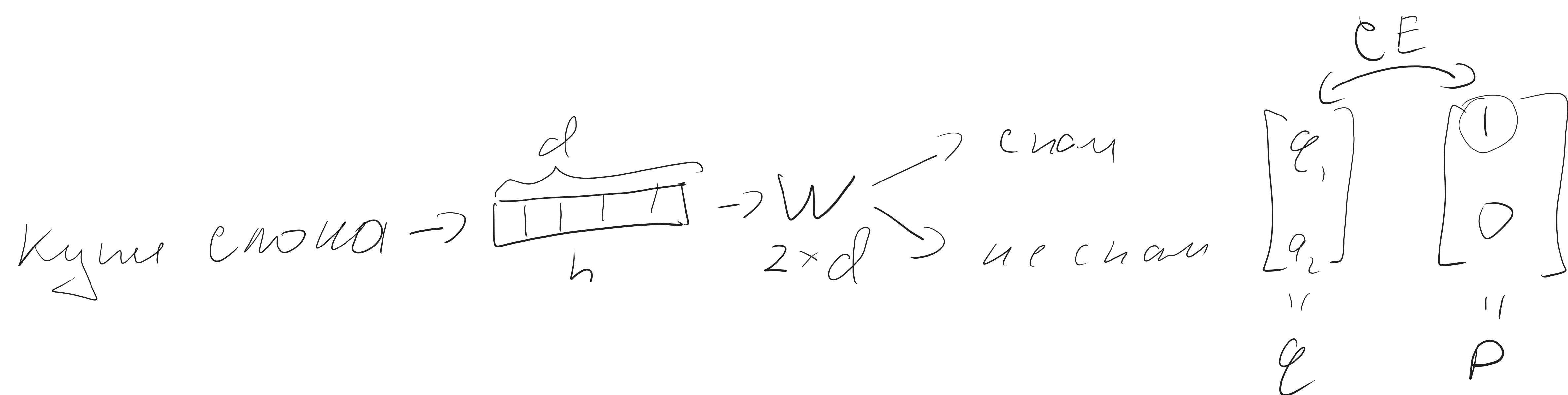


$$\frac{8}{2} = 4$$

$$\begin{cases} \text{Entropy} = p_0 \log\left(\frac{8}{2}\right) + p_1 \log\left(\frac{8}{2}\right) \\ \log(a^{-1}) = -\log(a) \end{cases} \quad + p_2 \log\left(\frac{8}{4}\right) = \mathbb{E}(\sum p_i \log(p_i))$$

$$\text{Cross-Entropy} = (p_0 \log(q_0) + p_1 \log(q_1) + \dots)$$

$$KL_{p||q} = \text{CE} - \text{Entropy}$$



$$\begin{aligned} \text{CE} &= (1 \cdot \log(q_1) + 0 \cdot \log(q_2)) = \\ &= -\log(q_1) \end{aligned}$$